

Karnataka Magnetic Inversion: Comparing Regularizations

The airborne magnetic data of the area of the gravity comparison, continued upwards and inverted at full resolution on AWS EC2, with the terrain in the mesh. Seven regularization settings are compared for data fit, lateral structure and depth, and six further runs test the choices made in preparing the data.

AREA	DATA
70 × 70 km, easting 641–711 km, northing 1634–1704 km (UTM 43N, EPSG:32643)	Total magnetic intensity anomaly, 37.5 m grid flown 80 m above the ground; continued upwards to 1 km above the ground and sampled at 1 km: 5,041 points; second-order trend surface removed
FIELD	MESH
IGRF 2020 at the centre: 42,100 nT, inclination 19.3°, declination –1.4° to grid north; induced magnetization only, susceptibility 0 to 1 SI	As for gravity with terrain: 1 km × 1 km × 250 m core below the Copernicus DEM, 84 × 84 × 50 = 352,800 cells, 335,518 below the ground
RUNS	
13 full-resolution inversions on AWS EC2 (ap-south-1, c5.4xlarge), 12.6–15.4 minutes each, \$2.11 in total	

Summary

- **The data have to be continued upwards before they are thinned.** As flown, 18% of the grid's variance is at wavelengths shorter than 2 km, which sampling at 1 km aliases; at 1 km above the ground 0.07% is left. Inverted as flown, the data cannot be fitted (χ^2/N 1.45, RMS 209 nT) and the model is smeared to depth.
- **The depth weighting decides where the susceptibility goes, and the textbook value fails.** With the Li & Oldenburg exponent $\beta = 3$ usually quoted for magnetic data, 82% of the susceptibility × volume lies below the 10 km core of the mesh; with $\beta = 2$, 70%; with sensitivity weighting, 63%. With $\beta = 0.5$ and 1 and with L1–L2 it is 8%–10%, and 62%–78% is in the core.
- **All seven fit the data equally** (χ^2/N 0.83–1.07), so the fit cannot choose. The centroid of the magnetic rock of the Sandur belt lies at 0.8 km for $\beta = 0.5$, 2.1 km for $\beta = 1$, 2.8 km for L1–L2, 2.9 km for $\beta = 1.5$ and 5.6 km for $\beta = 3$: in every case shallower than the dense body there (4.8 km in the gravity model with $\beta = 1$).
- **The residuals are not random.** Every model leaves the positive anomaly on the southern side of the Sandur belt underfitted by more than 150 nT. A model magnetized only along the present field does not explain the belt completely; remanent magnetization of the iron formation is the likely reason.
- **Susceptibility values are not rock values.** The compact models hold most magnetic cells at the 1 SI bound; with a bound of 0.3 SI the same data are fitted by a body 4.0 times the size. What the data constrain is the product: 120–228 SI·km³ in the Sandur belt for $\beta \leq 2$ and L1–L2.

- **The magnetic and the dense rock are related but not the same.** The magnetic sheets lie along the margins of the dense belt and inside it; column by column the two integrated models correlate at only 0.24.
- **Recommendation.** Continue the data upwards to about one cell above the ground; sparse with $\alpha_s = 1$ and depth weighting β between 0.5 and 1, with L1–L2 as an independent check; report the depth as a range and the susceptibility as a product with volume.

1 Data and preparation

1.1 Upward continuation

The survey (NGPM Block 8) was flown 80 m above the ground on lines 300 m apart; the grid has 37.5 m cells. In the area the anomaly ranges from $-7,402$ to $+14,973$ nT, with the largest values over the banded iron formation of the Sandur belt (Figure 1).

A mesh of 1 km cells cannot produce the short wavelengths such data contain, and sampling the grid at 1 km would alias them: 18% of the grid's variance in the area is at wavelengths shorter than 2 km. The grid was therefore continued upwards by 920 m, to 1 km above the ground, before sampling (`geoinv3d/methods/continuation.py`): this multiplies every wavelength λ by $\exp(-2\pi \cdot 920 \text{ m} / \lambda)$, which leaves 0.07% of the variance below 2 km (1.2% at 500 m). The continued field is what a survey flown 1 km above the ground would have measured; its receivers are placed there, 1138–2052 m above sea level. At the nodes it ranges from $-1,511$ to $+2,462$ nT. Section 4 tests this choice.

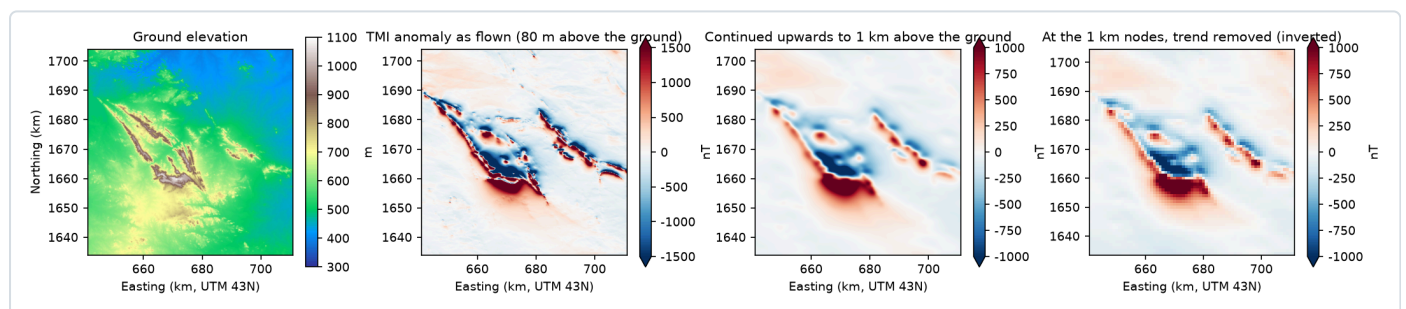


Figure 1 Left: ground elevation. Second: the magnetic anomaly as flown, 80 m above the ground. Third: continued upwards to 1 km above the ground. Right: the continued field at the 1 km nodes after removing a second-order trend surface, i.e. the data inverted.

1.2 Field, regional and errors

The inducing field is the IGRF 2020 field at the centre of the area: 42,100 nT, inclination 19.3°, declination -1.4° relative to grid north. At this low inclination a magnetized body gives a low over and north of it and a high to the south, which is the pattern of the Sandur belt in Figure 1. Only induced magnetization is modelled; the susceptibility is bounded to 0–1 SI.

A least-squares second-order trend surface (-105 to $+78$ nT) is removed as the regional field, as for gravity. Each datum has an error of 5% of its value plus 10 nT (10–132 nT, median 13 nT). The continued data are nearly noise-free; the error stands for what a model of 1 km cells with induced magnetization cannot reproduce. Section 4 tests a tighter error.

The mesh and the terrain are those of the gravity runs with terrain: cells above the DEM are left out, and the core reaches 10 km below the lowest ground. All depths are measured below the ground of each column.

1.3 The runs

Run	Regularization	Depth weighting	χ^2/N	RMS (nT)	Max residual (nT)	Iterations	EC2 time (min)
sparse, sensitivity weighting	sparse (IRLS), $\alpha_s = 1$, $p = [0, 2, 2, 2]$	sensitivity	0.93	41	618	46	14.5
sparse, $\beta = 0.5$	sparse (IRLS), $\alpha_s = 1$, $p = [0, 2, 2, 2]$	Li & Oldenburg, $\beta = 0.5$	0.91	53	750	45	13.9
sparse, $\beta = 1$	sparse (IRLS), $\alpha_s = 1$, $p = [0, 2, 2, 2]$	Li & Oldenburg, $\beta = 1$	0.90	49	680	45	13.9
sparse, $\beta = 1.5$	sparse (IRLS), $\alpha_s = 1$, $p = [0, 2, 2, 2]$	Li & Oldenburg, $\beta = 1.5$	0.83	46	678	44	13.8
sparse, $\beta = 2$	sparse (IRLS), $\alpha_s = 1$, $p = [0, 2, 2, 2]$	Li & Oldenburg, $\beta = 2$	1.03	46	710	44	14.8
sparse, $\beta = 3$	sparse (IRLS), $\alpha_s = 1$, $p = [0, 2, 2, 2]$	Li & Oldenburg, $\beta = 3$	0.87	41	718	44	15.4
L1–L2 (IRLS)	elastic net (Utsugi 2019), L1 share 0.8, solved by IRLS	sensitivity	1.07	60	967	47	12.6

Common settings: susceptibility 0–1 SI; β chosen automatically for $\chi^2 = N$; at most 60 iterations and 40 IRLS iterations; $\alpha_x = \alpha_y = \alpha_z = 1$ as length scales. Li & Oldenburg depth weighting weights the model norm by $(z + z_0)^{-\beta/2}$, z the depth below the nearest receiver; $\beta = 3$ is the value usually quoted for magnetic data (the kernel decays as r^{-3}), $\beta = 2$ for gravity.

2 Comparing the seven settings

2.1 Data fit

All seven runs reach the target, χ^2/N 0.83–1.07, with a residual RMS of 41–60 nT (table above). The residual maps (Figure 2) are alike, and they are not noise:

- **The high south of the Sandur belt is underfitted in every model:** the observed field exceeds the predicted one by more than 150 nT at 57–113 nodes (1 km² each), most of them there, and falls short of it by as much at only 0–3. The error there is large (5% of up to 2,462 nT), so these residuals are only a few standard deviations: for $\beta = 1$ the normalized residuals have a 99th percentile of 3.8, 2.0% of them exceed 3, and the worst 1% carry 24% of χ^2 .
- **The same sign along the north-eastern belt** and at the north-western end of the Sandur belt: positive residuals on the belts, in all seven models.
- **What this means.** A body magnetized along the present field gives a fixed ratio between its low and its high. The observed high is stronger than any of the models can make it while fitting the low, which points to a magnetization direction other than the present field: remanent magnetization, common in banded iron formation. The models are therefore a first approximation in the belts.

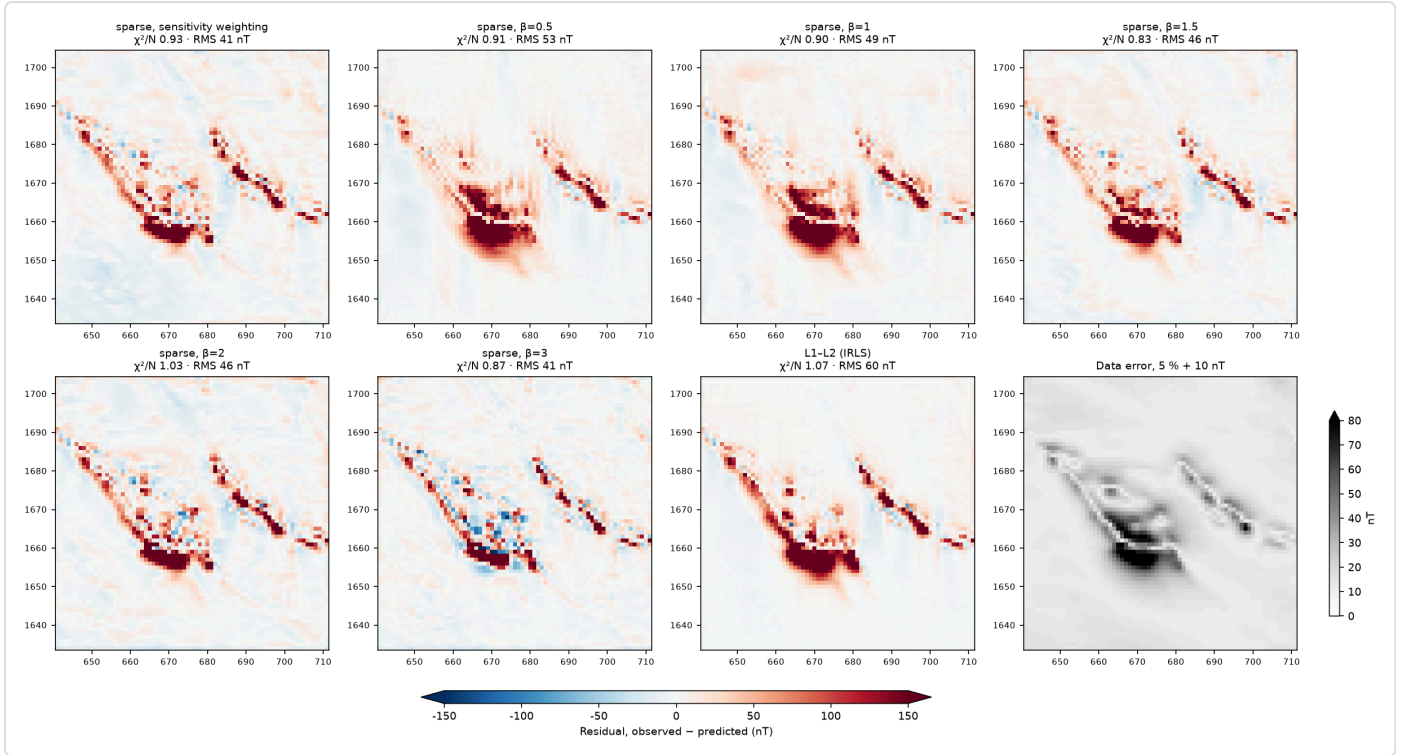


Figure 2 Residuals (observed – predicted) of the seven models, colour scale ± 150 nT, and the data error (bottom right).

2.2 Where the susceptibility goes

The table gives where the susceptibility $\times$ volume lies in each model, and measures of the magnetic rock in the Sandur belt (a box of 20×18 km around it, easting 658–678 km, northing 1657–1675 km). Figure 3 shows the distribution with depth.

Run	Core	Lateral padding	Below the core	D50 km	D90 km	Sandur belt: centroid km	10–90% depth range km	susceptibility $\times$ volume (SI $\cdot$ km ³)	largest value (SI)
sparse, sensitivity weighting	12%	25%	63%	10.8	15.2	3.4	1.1–6.1	209	1.00
sparse, $\beta = 0.5$	78%	14%	8%	0.9	6.4	0.8	0.1–1.4	120	1.00
sparse, $\beta = 1$	71%	20%	10%	3.4	7.6	2.1	0.6–3.9	152	1.00
sparse, $\beta = 1.5$	43%	12%	45%	5.9	11.9	2.9	1.1–5.1	187	1.00
sparse, $\beta = 2$	27%	4%	70%	10.1	15.2	3.7	1.4–6.4	228	1.00
sparse, $\beta = 3$	17%	1%	82%	12.5	15.2	5.6	2.1–9.4	392	1.00
L1–L2 (IRLS)	62%	28%	10%	4.1	9.6	2.8	0.6–5.4	161	0.68

Core, lateral padding and below the core add up to 100%. D50, D90: half and 90% of the susceptibility $\times$ volume under the core area lie above this depth below the ground. Red: core $< 50\%$, lateral padding $> 30\%$, below the core $> 30\%$, D50 > 7 km.

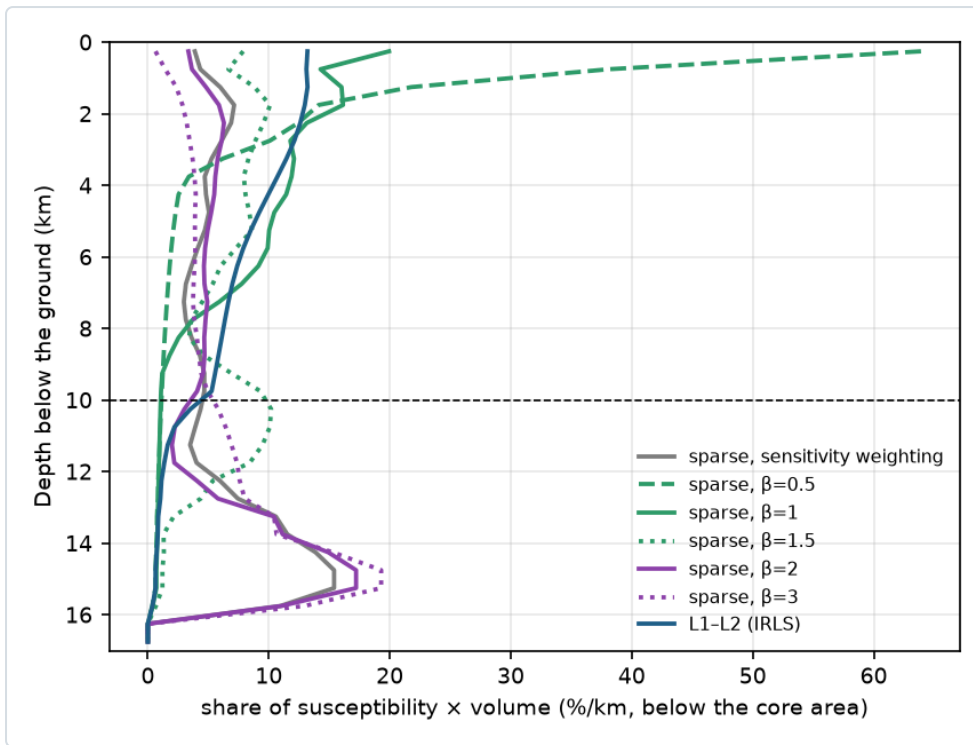


Figure 3 Distribution of susceptibility × volume with depth below the ground, under the core area (percent per km). The dashed line is 10 km, the depth of the core.

- **$\beta = 2$, $\beta = 3$ and sensitivity weighting fill the bottom of the mesh.** They leave 70%, 82% and 63% of the susceptibility × volume below the core, in the large padding cells, where the data cannot tell what is there. These weightings make deep cells cheap, and the data, fitted within 5%, do not object.
- **$\beta = 0.5$, $\beta = 1$ and L1–L2 keep it in the core:** 78%, 71% and 62%, with 8%, 10% and 10% below it. $\beta = 1.5$ is in between (45% below the core).
- **About a fifth to a quarter lies in the lateral padding** for the shallow settings. The belts leave the area at its north-western and south-eastern edges, and their anomalies are cut there.
- **For gravity the same exponents behaved differently:** $\beta = 1$ gave a main body centred at 5.4 km, and only $\beta = 1.5$ began to trail to depth. For both methods the usable exponents are well below the textbook values of 2 and 3.

2.3 The Sandur belt in section

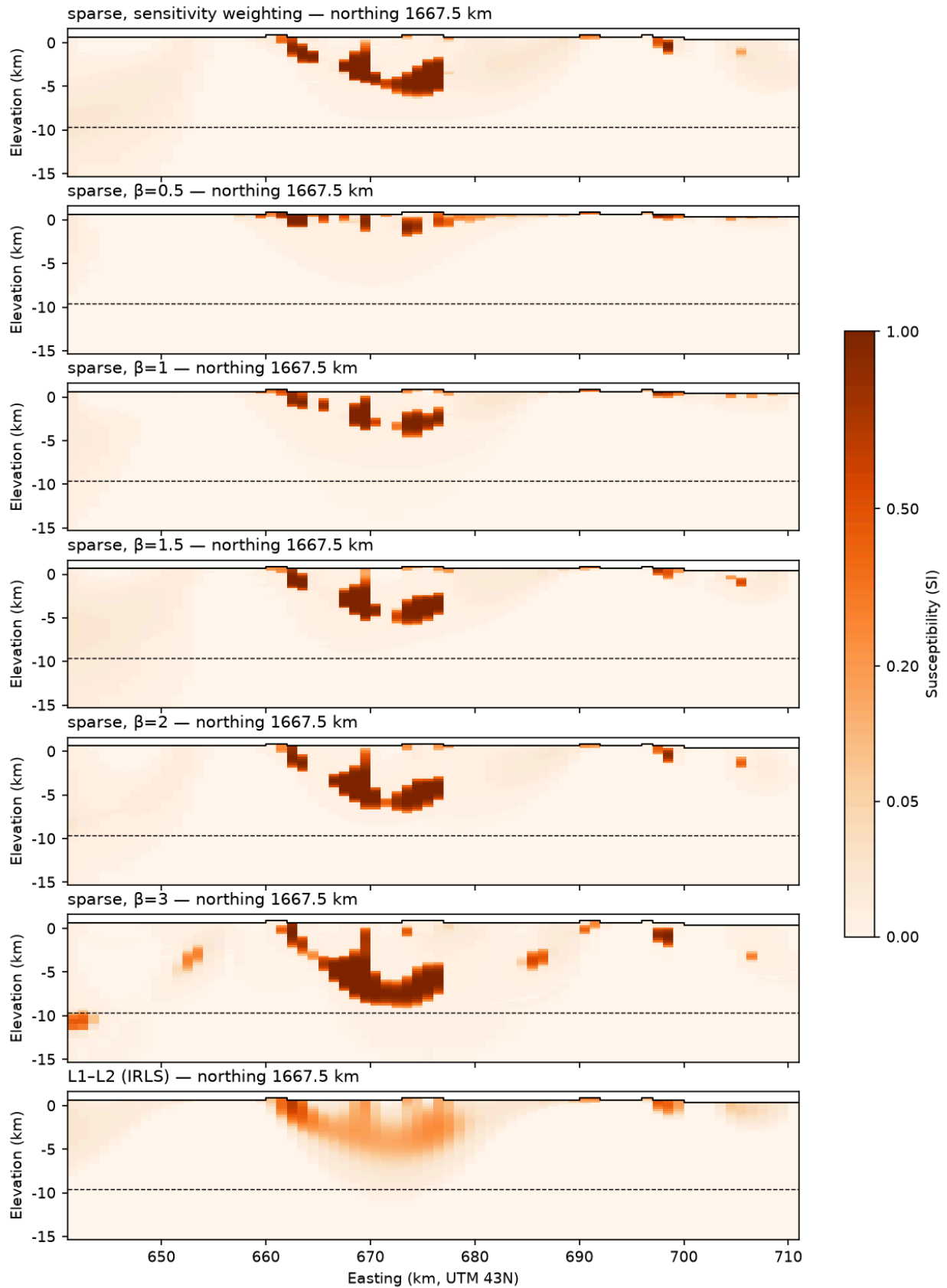


Figure 4 E–W sections through the Sandur belt (northing 1667.5 km) in the seven models. The black line is the ground of the mesh, the dashed line the base of the core. The colour scale is a square-root scale from 0 to the bound of 1 SI.

- **The sparse models are made of cells at the bound.** The largest value in the belt is 1.00 SI in all six; the models are sheets and blocks one to three cells wide.

- **β sets the depth monotonically.** The centroid of the belt's magnetic rock lies at 0.8, 2.1, 2.9, 3.7 and 5.6 km for $\beta = 0.5, 1, 1.5, 2$ and 3 ; 80% of it lies between 0.6 and 3.9 km for $\beta = 1$ and between 2.1 and 9.4 km for $\beta = 3$.
- **A deeper model needs more magnetic rock:** 120, 152, 187, 228 and 392 SI·km³ in the box.
- **L1–L2 is smooth and does not reach the bound** (largest value 0.68 SI): a bowl from the surface to about 5 km with its centroid at 2.8 km, between $\beta = 1$ and $\beta = 1.5$, and 161 SI·km³.
- **The shape.** In every model the magnetic rock of the belt forms a bowl that is open upwards, with sheets reaching the surface at the ridges on both sides. This is the shape of a synform with iron formation on its limbs, which is how the Sandur belt is mapped; but the bowl also resembles the way a compact inversion closes a body at depth, so the base of the bowl is no more certain than its depth.

2.4 Lateral structure

Figure 5 shows five models at about 1, 3 and 6 km below the mean ground, Figure 6 the vertically integrated susceptibility and the centroid depth of each column.

- **The belts are in the same place in every model:** the two limbs of the Sandur belt, its closure in the south-east and the parallel north-eastern belt.
- **Cell by cell the models differ more than the gravity models did.** The sheets are one or two cells wide and move by a cell from one setting to the next: the integrated maps of $\beta = 0.5, \beta = 1$ and L1–L2 correlate at 0.66–0.78 as they are and at 0.89–0.95 after smoothing over 5 km (table below). The deep settings differ from these also at 5 km.
- **Bodies at the corners of the area** (south-west, south-east, north) appear in every model. They fit anomalies that the edge of the area cuts and should not be interpreted.
- **North–south sheets** north of the Sandur belt (near easting 652, 660 and 683 km) need caution. At an inclination of 19° a body elongated along magnetic north gives a weak anomaly except at its ends, so its length is poorly constrained, and a compact inversion tends to stretch bodies in that direction.

	sens.	$\beta = 0.5$	$\beta = 1$	$\beta = 1.5$	$\beta = 2$	$\beta = 3$	L1–L2
sens.	–	0.48	0.60	0.61	0.67	0.69	0.58
$\beta = 0.5$	0.48	–	0.90	0.64	0.50	0.41	0.95
$\beta = 1$	0.60	0.90	–	0.84	0.67	0.49	0.89
$\beta = 1.5$	0.61	0.64	0.84	–	0.88	0.50	0.68
$\beta = 2$	0.67	0.50	0.67	0.88	–	0.59	0.59
$\beta = 3$	0.69	0.41	0.49	0.50	0.59	–	0.50
L1–L2	0.58	0.95	0.89	0.68	0.59	0.50	–

Correlation between the vertically integrated susceptibility maps in the 70 × 70 km core, smoothed over 5 km. Green: ≥ 0.85.

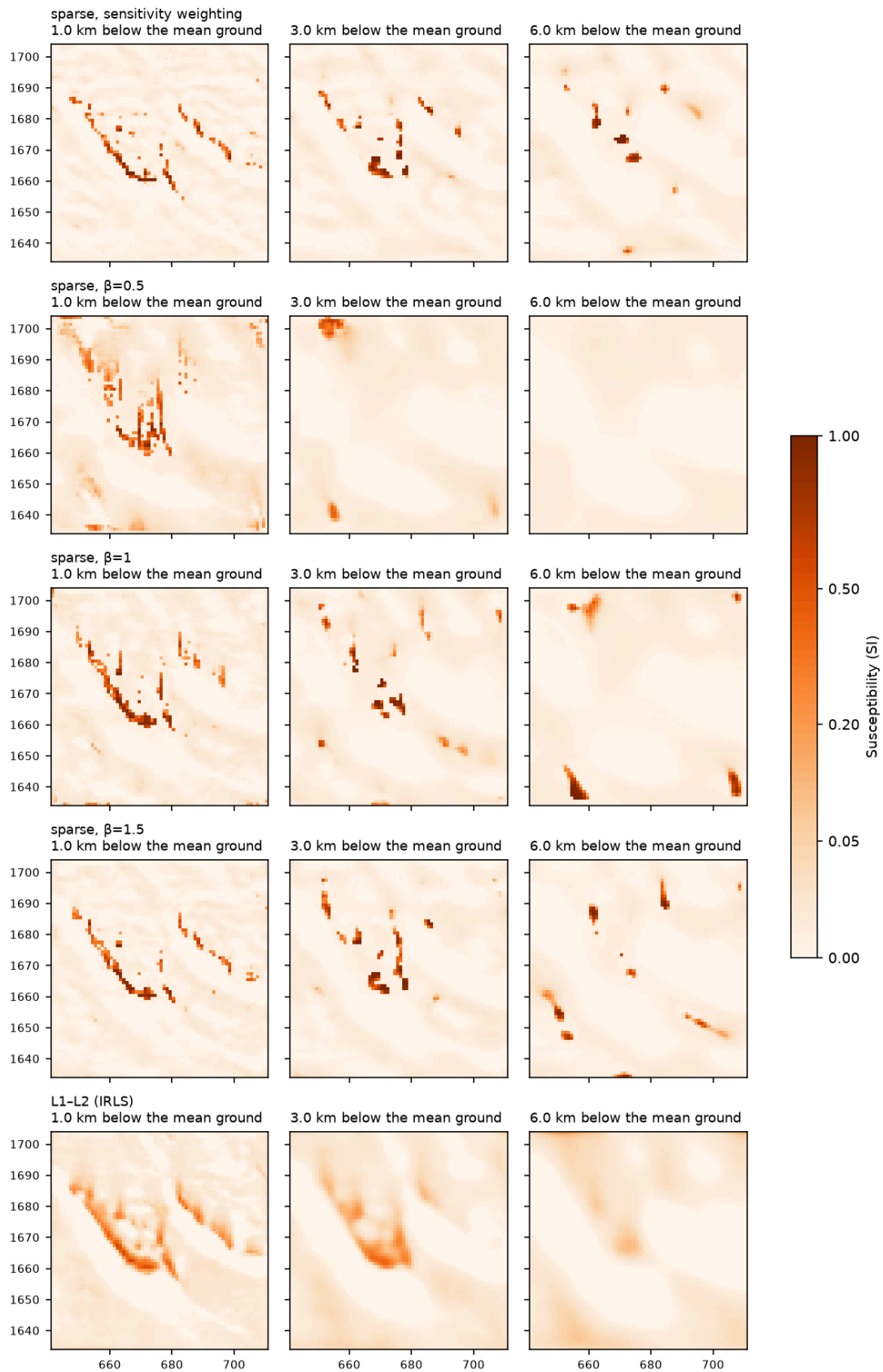


Figure 5 Layers of five models at about 1, 3 and 6 km below the mean ground (core area only); square-root colour scale, 0 to 1 SI.

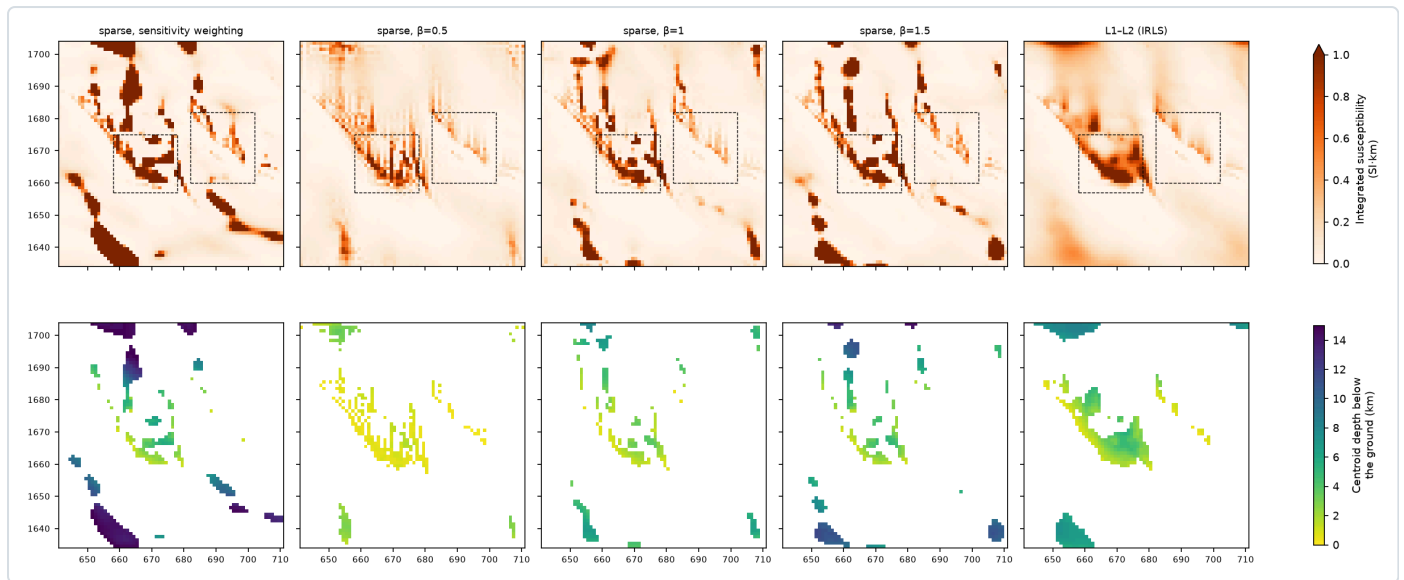


Figure 6 Top: vertically integrated susceptibility (SI·km); the dashed boxes are the Sandur belt and the north-eastern belt. Bottom: centroid depth of the susceptibility below the ground in each column (only columns whose integral exceeds 20% of the maximum).

3 Cross-check: the synthetic test

On 28 September the same settings were tested on synthetic data with known bodies under the same inducing field: a shallow block (1–3 km), a deep block (4–7 km) and a dyke dipping 45°, on a flat earth, with the data 80 m above it on a 1 km grid (`examples/synthetic_magnetic_karnataka.py`). Depths are centroids of the recovered susceptibility in each body's footprint.

Run (synthetic test, 28 September)	χ^2/N	Correlation with the true model	Shallow block, true 2.0 km	Deep block, true 5.5 km	Dyke dip, true 45°
sparse, sensitivity weighting	1.11	0.21	3.9 km	8.5 km	64°
sparse, $\beta = 2$	1.07	0.24	3.7 km	8.0 km	62°
sparse, $\beta = 3$	1.31	0.17	4.9 km	8.6 km	66°
L1–L2 (IRLS)	1.00	0.60	3.9 km	6.7 km	65°

- **Sensitivity weighting, $\beta = 2$ and $\beta = 3$ put the bodies 1.5–3 km too deep** and recovered little of them; $\beta = 3$ was the deepest. On the field data the same three settings are the ones that fill the bottom of the mesh.
- **L1–L2 was the best of the four** (correlation 0.60 with the true model, the deep block at 6.7 km for 5.5). On the field data it agrees with $\beta = 1$ at 5 km resolution (correlation 0.89).
- **Smaller exponents were tried only on a coarse mesh in that test:** there $\beta = 1$ put the deep block at its true depth. The field-data runs support the same choice, and a full-resolution synthetic test with $\beta = 0.5$ and 1 would confirm it.

4 Tests of the set-up

Six runs change one thing at a time from the run with $\beta = 1$ (Figure 7):

Run (sparse, $\beta = 1$)	χ^2/N	RMS (nT)	Core	Below the core	Sandur belt: centroid km	susceptibility × volume (SI·km ³)	largest value (SI)	Integrated susceptibility at 5 km: correlation with the reference
Reference ($\beta = 1$)	0.90	49	71%	10%	2.1	152	1.00	1.00
$\alpha_s = 0.1$	1.00	52	66%	12%	2.1	150	0.81	0.98
Upper bound 0.3 SI	0.94	51	66%	15%	2.5	173	0.30	0.90

Run (sparse, $\beta = 1$)	χ^2/N	RMS (nT)	Core	Below the core	Sandur belt: centroid km	susceptibility $\times$ volume (SI $\cdot$ km ³)	largest value (SI)	Integrated susceptibility at 5 km: correlation with the reference
Error 2% + 5 nT	1.08	16	50%	28%	2.1	179	1.00	0.73
Continued to 500 m above the ground	1.00	69	61%	16%	2.1	161	1.00	0.95
As flown, 80 m above the ground (not continued)	1.45	209	30%	38%	4.3	366	1.00	0.35
Flat earth	1.02	51	68%	12%	2.2	149	1.00	0.96

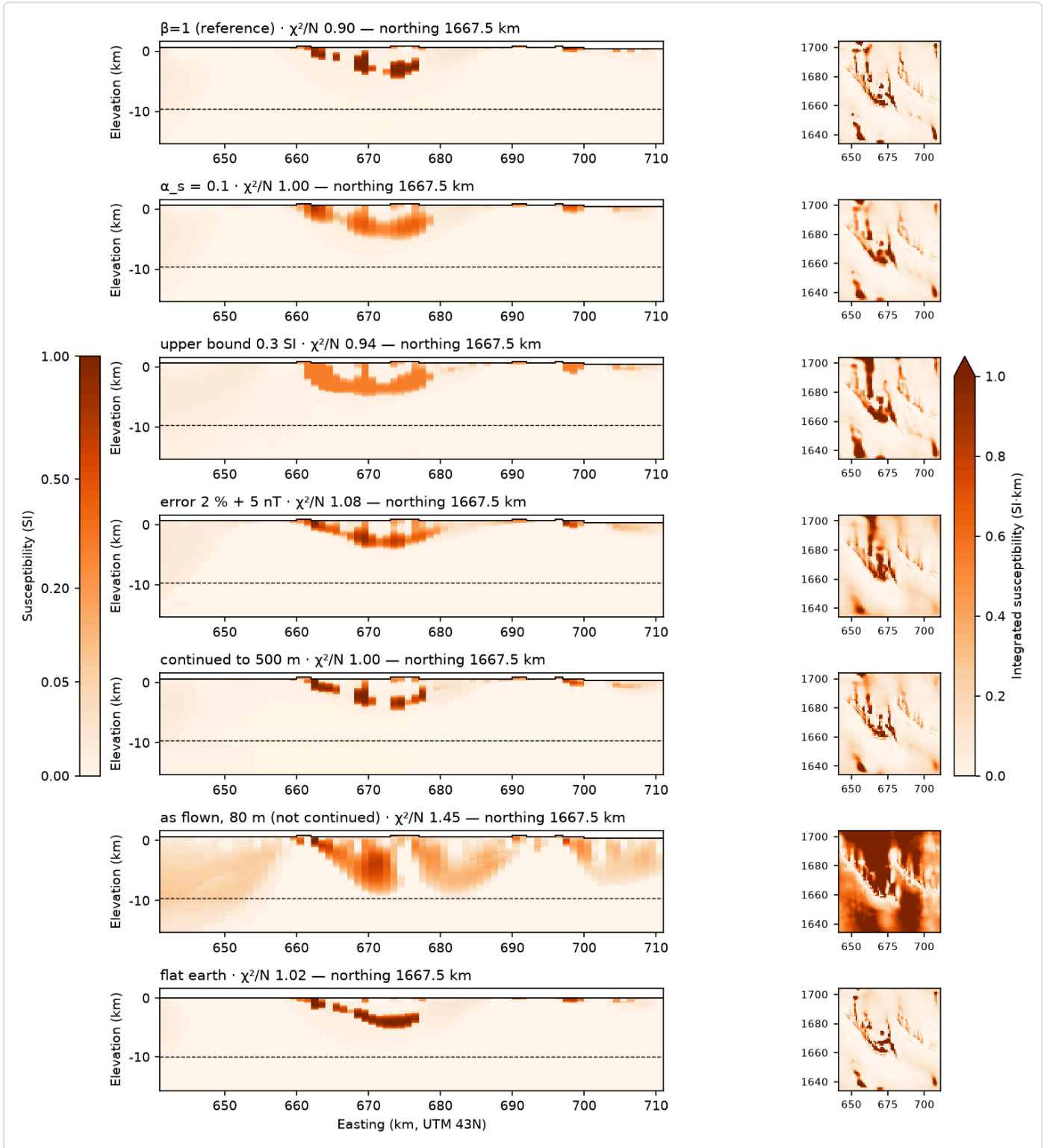


Figure 7 The reference run and the six tests. Left: E–W sections through the Sandur belt (square-root colour scale, 0 to 1 SI). Right: vertically integrated susceptibility.

- $\alpha_s = 0.1$ gives a smoother model that stays below the bound (largest value 0.81 SI), with the same depth and the same susceptibility $\times$ volume.
- **An upper bound of 0.3 SI** gives nearly the same product (173 against 152 SI $\cdot$ km³) in a larger body (583 against 144 km³ above half the largest value), 0.5 km deeper. The data fix the product, the bound fixes the volume. The two iron-formation samples of the area have 0.05 SI, far less than either bound: the magnetic rock is not represented by the hand samples, or fills only a part of each cell.
- **An error of 2% + 5 nT** can be fitted (χ^2/N 1.08, RMS 16 nT), with more detail in the belt and more susceptibility below the core (28% against 10%). The residual high south of the belt remains, at a lower level.
- **Continuing to 500 m instead of 1 km** gives the same model at 5 km resolution (correlation 0.95) and a rougher fit (RMS 69 nT).
- **Without continuation** the inversion does not reach the target in 55 iterations (χ^2/N 1.45, largest residual 9,826 nT) and spreads weak susceptibility through the whole mesh: the aliased data are not the field of any model on this mesh.
- **A flat earth** (receivers 1 km above a plane) changes little: the centroid of the belt is at 2.2 km against 2.1 km, the correlation at 5 km is 0.96. At 1 km above the ground the relief of a few hundred metres matters little for the magnetic data; it matters for the joint inversion, where both models must share the mesh of the gravity data.

5 Magnetism and gravity

Figure 8 compares the magnetic model with the gravity model of the same area (both sparse with $\beta = 1$, with terrain).

- **The dense belt and the magnetic belt are the same structure.** The Sandur belt is the main body of both models; at 5 km resolution the strongest magnetic rock lies within the south-eastern half of the dense belt, and the north-eastern magnetic belt runs beside a smaller dense belt.
- **Inside the belt they are different rock.** The magnetic sheets follow the margins of the dense body and its south-eastern closure; the dense rock fills the belt. Column by column the integrated susceptibility and the integrated density correlate at 0.24; 31% of the most magnetic fifth of the columns are dense (more than 0.3 g/cc $\cdot$ km), and 44% of the dense columns are in that fifth.
- **They are at different depths.** The magnetic rock of the belt is centred at 2.1 km, the dense rock at 4.8 km (both with $\beta = 1$), and both depths follow the regularization.

This is what a greenstone belt with iron formation should look like: thin, strongly magnetic iron formation on the limbs, and a thick pile of dense, weakly magnetic metavolcanic rock in the core (the amphibolite and metabasalt samples: 2.9–3.0 g/cc, 0.0002–0.004 SI). For a joint inversion it means that the two models share boundaries rather than values, which favours a structural coupling; the joint inversion of these two datasets is the subject of a separate report.

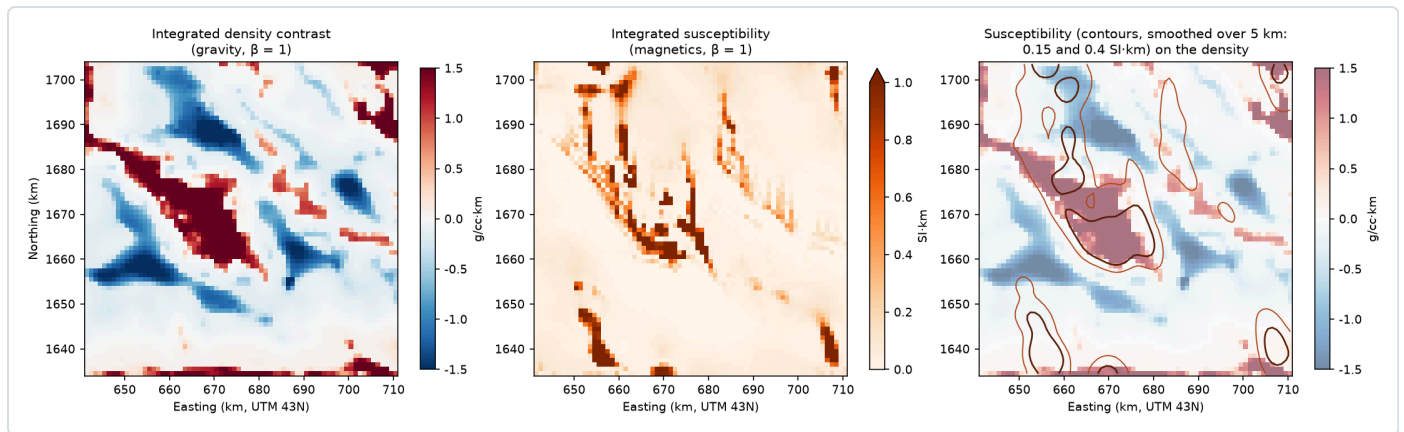


Figure 8 Left: vertically integrated density contrast of the gravity model with terrain ($\beta = 1$). Centre: vertically integrated susceptibility ($\beta = 1$). Right: the susceptibility, smoothed over 5 km, as contours on the density.

6 Recommendations

- **Preparation:** continue gridded magnetic data upwards to about one cell size above the ground before thinning them, and place the receivers there. Thinning alone is not enough.
- **Settings:** sparse, $\alpha_s = 1$, $p = [0, 2, 2, 2]$, depth weighting with β between 0.5 and 1; L1–L2 as an independent check. Not $\beta = 3$, not $\beta = 2$, and not sensitivity weighting on a mesh with deep padding.
- **Report ranges:** the magnetic rock of the Sandur belt is centred 0.8–2.8 km below the ground in the three recommended models and holds 120–161 SI·km³. Do not read the cell values as rock susceptibilities.
- **Remanence:** the systematic residuals call for a magnetization-vector inversion, or for an amplitude quantity that does not depend on the direction, in the belts.
- **Depth needs independent constraints**, as for gravity: susceptibility logs or samples of the fresh iron formation, its mapped dips, or the gravity data through a joint inversion.

Files

- Inputs (DEM, continued magnetic data at the nodes): `examples/output/karnataka_inputs/` (`prepare_inputs.py`)
- This report and its figures: `examples/output/karnataka_magnetic/scripts/` (`make_figures.py`, `build_report.py`)
- The 13 full-resolution results: `data/`; launched with `deploy/ec2_multi_run.py karnataka-magnetic`
- All runs in one interactive workflow (DAG viewer): `karnataka_magnetic.geoinv3d_viewer.html`
- The gravity comparison with terrain: `examples/output/karnataka_gravity_terrain/`